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We know who wins: graph-oriented approaches of passing networks for predictive football match outcomes

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Abstract

Football is one of the most popular sports worldwide. Football fans, in their pursuit of maximizing enjoyment from supporting their favorite teams, are eager to know the outcome of ongoing matches. However, related studies to predict football match outcomes have focused on predicting them before the match starts, using measures from previous match data. Therefore, we propose a method to predict the outcome of an ongoing match at any time during the game as one of three classes: home win, away win, or draw. We focus on the event of passes, the most fundamental unit of movement in football, and have been shown in many studies to influence match outcomes. Unlike previous studies that used generalized aggregated variables such as total pass counts for pass information, we propose a graph-based classification model that directly utilizes the dynamically changing passing network as a graph. Our proposed graph-based model effectively leverages information about the location of each pass and the players attempting these passes from the passing network. As a result, in experiments predicting the final outcome at the 45, 60, 75, and 90-minute marks, our model outperformed baseline models that used general pass measures. Furthermore, by additionally incorporating non-pass event information, our model achieved a performance improvement of 5 to 20% across various classes compared to using only the passing network. This study holds substantial implications for integrating graph theory and graph modeling into the domain of football, highlighting the impact of passing information on outcome prediction.

Keywords: Graph approach, Passing network, Match outcomes, Football, Soccer

Introduction

Football is one of the most popular sports globally, boasting the largest market size among all sports. According to *Statista*,¹ the revenue of the five major football leagues (English, Spanish, German, Italian, and French) in the 2022/23 season was projected to exceed 2 billion euros each, with the English Premier League (EPL) alone expected

¹ <https://www.statista.com/statistics/261218/big-five-european-soccer-leagues-revenue/>



Fig. 1 An example of the 'Win Probability' feature during the live broadcast of the English Premier League

to reach 6.6 billion euros. This substantial economic footprint underscores the global importance of the sport.

What makes people so passionate about football? Its appeal extends beyond the field, rooted in emotional and social connections. According to Tokuyama et al. [1], four key motivations drive football viewership: *affiliation, aesthetics, competition, and social facilitation*. The *competition* variable, in particular, plays a crucial role in engaging audiences. Martins et al. [2] further emphasized that witnessing one's supported team win is a significant factor influencing viewership. Their findings suggest that uncertainty regarding a match's outcome (outcome uncertainty) may negatively impact audience retention. This highlights the necessity of *real-time prediction models*, which can enhance engagement by providing up-to-the-minute insights into match progress.

To address outcome uncertainty, predictive insights have been integrated into football broadcasts. For example, the EPL introduced the 'Win Probability' feature in its live broadcasts during the 2021/22 season, powered by *Oracle Cloud*. This feature visually represents the likelihood of each team winning, based on a computational model trained on a four-year dataset comprising approximately 100,000 match instances.²

The integration of machine learning and deep learning models into football analytics has grown significantly in recent years. While many studies have explored predictive models for match outcomes [3–5], most have focused on pre-match forecasts rather than in-game, real-time predictions. However, viewers, analysts, and coaching staff are particularly interested in *real-time fluctuations in winning probabilities*, which dynamically change based on match events. Real-time prediction models serve as valuable tools for understanding the game as it unfolds, providing precise insights into tactical shifts and performance trends. Coaches can use these models to make informed decisions,

² <https://www.premierleague.com/news/2210969>

viewers gain a deeper appreciation for in-game developments, and commentators can offer more data-driven narratives about the match (Fig. 1).

Among all match events, *passing* is the most fundamental and frequent action, shaping match outcomes. According to Cintia et al. [6], approximately 80% of all events in a football match involve passing. This has led researchers to analyze team strategies and performance through passing patterns [7–9]. Given the significance of passing, the most intuitive method for representing team passing behavior is through *passing networks*. Figure 2 illustrates an example, where nodes represent players, and edges denote passes. The size of each player node is proportional to their passing frequency, while edge thickness indicates the volume of passes exchanged. Passing networks provide a dynamic framework for analyzing team performance and playing style [10, 11].

While passing-related data has been integrated into previous predictive models, direct use of passing networks for real-time match outcome prediction remains largely unexplored. Instead, most studies have utilized passing-related information as one of many variables input into their machine learning models for pre-match prediction, often aggregating passing data into general statistical measures [12–14]. This approach limits the ability to capture the unique characteristics of each pass, as tabular representations struggle to retain the positional and directional information inherent in passing sequences. For example, commonly used features such as total pass count or average pass accuracy fail to reflect the dynamic and structural relationships within passing networks. Furthermore, understanding how frequently one player passes to another is crucial for modeling a team's playing style, yet most models fail to integrate this information effectively.

Thus, our research question is:

- Can we predict the outcomes of ongoing football matches using passing networks, with a focus on real-time prediction?

To address this question, we developed a graph classification model based on Graph Attention Networks (GATs) [15]. The model utilizes passing graphs of both the home and the away teams as input and is trained to classify match outcomes into three distinct classes: Home Win, Away Win, and Draw. In addition, we enhance the model's predictive capability by integrating non-pass events, allowing for a more

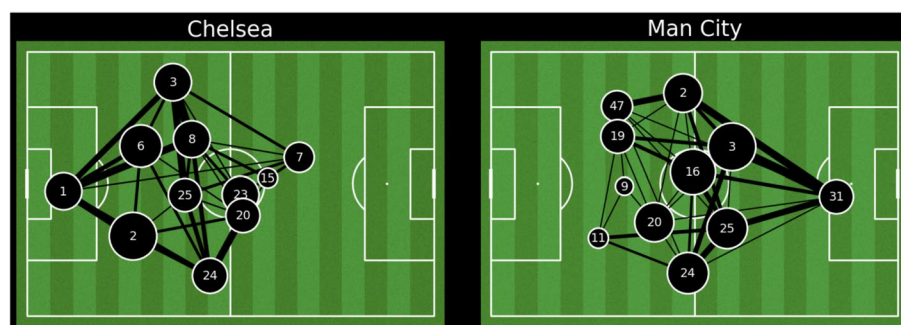


Fig. 2 An instance of the passing network from the match between Chelsea and Manchester City (November 13, 2023). The plotted nodes represent the initial lineups for both teams, with player positions shown as their average locations on the pitch

comprehensive assessment of real-time match developments. Consequently, our model provides continuous predictions of home win/away win/draw probabilities at any point during a match, offering a more dynamic and adaptive approach to match forecasting.

While several prior research explored the use of passing networks and graph-based learning, their goals and methodologies differ from our approach. Beheshtian-Ardakani et al. [16] introduced a Complex Multiplex Passing Network (CMPN) model, which analyzed passing interactions across multiple layers to assess team strategies at different topological scales. Their research primarily focused on using network-based indicators for attacking play analysis rather than real-time match outcome prediction. Similarly, Rahimian et al. [17] employed a Temporal Graph Network (TGN) framework to model passing dynamics, estimating Receiver Selection Probability (RSP) and Receiver Prediction Probability (RPP) to evaluate pass selection tendencies. Although they highlighted the growing relevance of graph-based models in football analytics, they do not focus on real-time match forecasting using GATs. Our study distinguishes itself by integrating passing networks into a GAT-based model, designed to analyze the evolving team-wide dynamics and continuously predict match outcomes in real-time.

The structure of the paper is organized as follows: "Related works" section introduces previous studies aligned with similar research goals or methodologies. In "Method" section, we state the process of collecting and preprocessing the data used in this study, elucidating the construction of passing graphs and the architecture of our graph classification model. "Empirical analysis" section encapsulates the classification performance results obtained from empirical experiments. We further analyze the utility of passing networks and graph modeling by thoroughly taking graph embeddings. Finally, in "Conclusion" section, we conclude our study by highlighting new insights and suggestions gleaned from this research endeavor.

Related works

Predicting football outcomes

Earlier studies on football match outcome prediction primarily relied on statistical models or data distributions to anticipate match results or scores. For example, Dixon and Coles [18] used bi-variate Poisson distributions, while Rue and Salvesen [19] employed Bayesian generalized linear models. Goddard [20] conducted a comparative analysis of score prediction versus result prediction models. Hvattum [21] adapted ELO ratings for each team and demonstrated improved predictive performance in betting scenarios. Carpita et al. [22] applied data mining techniques to identify match outcome determinants in the Italian Serie A, using Random Forest feature selection and Principal Component Analysis (PCA) over a multi-season dataset. These studies established fundamental statistical modeling approaches and stable long-term predictors of football outcomes.

Baboota and Kaur [5] developed position-based player rating systems and included match statistics such as shots taken. Their work highlights how contextual information can enrich prediction models. Wheatcroft [23] proposed a hybrid framework combining observed and forecasted pre-match statistics, yielding practical results in betting

applications. Rodriguez and Pinto [24] integrated team and player-level information across different model types, selecting Random Forest as the most effective based on overall performance. Stubinger et al. [25] introduced features derived from physical, in-game, and mental characteristics to improve outcome prediction.

With the rise of deep learning, more recent work has explored neural architectures for football analytics. Rahman [26] used neural network-based feature embeddings as input to an LSTM model to capture temporal dynamics in football data. However, while deep learning methods are increasingly used for tasks like action recognition-e.g., passing, shooting, and event detection [27–29]-their use in match outcome prediction remains relatively limited.

In parallel, expected goals (xG) modeling has emerged as a popular framework for offensive evaluation. Carpita et al. [30] introduced role-based performance indicators by aggregating individual player statistics into latent factors using the Cluster of Variables Around Latent Variables (CLV) method. Cefis and Carpita [31] proposed an interpretable xG model that emphasized explainability across multiple classifiers, demonstrating the relevance of explainable AI (xAI) in sports analytics. Their work underscores a broader shift toward interpretable and context-aware predictive modeling.

Network theory and football passes

Applying graph analysis and modeling to football generally entails utilizing passing graphs. Grund [32] was among the first to apply network theory to football performance analysis, investigating the relationship between within-team network structures and team success in the English Premier League. By analyzing over 283,000 passes from 760 matches, the study demonstrated that teams with denser passing networks tend to perform better, while highly centralized structures negatively impact performance. Other initial studies focused on comparing teams using various metrics derived from these networks. Lopez et al. [7], for instance, scrutinized network measures (like connectivity, clique, clustering coefficients, etc.) of teams that reached the round of 16 in the 2010 FIFA World Cup, and revealed that the Spanish displayed the highest clustering coefficient and significant edge connectivity.

Yamamoto and Yokohama [33] perceived passing networks as dynamic systems to assess team performance and analyzed triangles formed in 5-minute intervals to discover the correlation between their counts and successful attacks. Cintia et al. [6] demonstrated that the mean and variance degree of nodes within a team's passing network correlated with its success. Buldu et al. [34] could be presented as one of the seminal studies applying graph analysis to football. They analyzed how Pep Guardiola's FC Barcelona, differentiated from other teams by comparing graph metrics, and discovered statistically significant differences in these values. If Buldu's research focused on a single team, Mclean's study [35] concentrated on a single tournament, the 2016 UEFA European Championship (Euro 2016). They analyzed the Goal Scoring Passing Networks (GSPN) of all matches in Euro 2016 and discovered a commonality among these networks: low density and cohesion. Ievoli et al. [36] examined the role of passing network properties and performance indicators in predicting match outcomes. Their study explored the use of binomial logistic regression (BLR) and machine learning models to assess the

probability of winning a match, integrating passing network measures with traditional match-specific covariates such as fouls committed, distance covered, and shot attempts. This demonstrated how passing networks enhance predictive accuracy, reinforcing the importance of structural team interactions in football analytics.

Gama et al. [37], utilized a set of network analyses to identify key players and understand their impact on overall team performance. Their intra-team level analysis highlighted left defenders assuming dominant roles during attacking phases. Gonçalves et al. [38] compared passing networks of Portugal's U-15 and U-17 association football teams and found that the U-17 team's network exhibited lower dependence on specific players and higher intra-team cohesion. Cartas et al. [39] constructed graphs with players, referees, and goalkeepers from soccer videos as nodes and utilized the network for player identification and action-spotting tasks. Zhou et al. [40] suggested a new way of weighting passing networks by the endpoint coordinates of each pass and found the level of consistency between those metrics and teams' overall performance. More recent work has begun to emphasize the critical influence of passing sequences themselves. Cao [41] is one such recent example. He investigated how the path of the ball prior to a shot influences shooting outcomes, using a dataset of over 30,000 shots from elite competitions. By extracting and analyzing eighteen passing path features, the study demonstrated that the trajectory of passes leading up to a shot plays a critical role in determining its success—highlighting the predictive power of passing sequences beyond mere positional data, highlighting how the trajectory of passes leading up to a shot can significantly affect its success.

While these studies demonstrate the growing use of network-based methods in football analytics, there remains a noticeable gap in the literature concerning real-time or ongoing prediction of match outcomes using dynamic passing network data. Moreover, many existing approaches rely on descriptive statistics or handcrafted features, with limited exploration of graph-oriented learning models capable of processing the structural complexity of passing networks. This gap motivates our proposed method, which leverages graph neural networks for real-time outcome prediction based on the evolving structure of team interactions.

Method

In this section, we delve into the intricacies of our data collection methodology and provide a comprehensive overview of the steps involved in preprocessing the acquired data and constructing passing network graphs. In addition, we introduce the structure of a total of three variations of the proposed graph-based classification model structure that takes each team's passing network as input.

Data collection

The dataset utilized in this study encompasses all league matches from both the EPL and the English Championship League across the 20/21, 21/22 and 22/23 seasons, comprising a total of 2835 games (later excluded 27 matches with data errors). Among

these, there are 1,200 games classified as Home Win, 905 games classified as Away Win, and 703 games classified as Draw. The data was gathered through direct scraping from *whoscored.com*,³ a website that presents football match details and events occurring within each game. Each match has its dedicated web page, and the pertinent match information is embedded within the HTML source code in JSON format. Leveraging regular expressions, we accessed and downloaded the HTML files of all EPL and Championship game web pages for the specified seasons, allowing us to extract the embedded JSON files containing the match information. Not only the EPL, but also all leagues for which match data is provided on *whoscored.com* (including the top 5 European leagues, lower leagues in the UK, Champions League, and other European leagues such as those in the Netherlands and Portugal) can be obtained using the same method.

To conduct our research, we required data containing information about all events as well as players during football matches. Finding publicly available datasets that clearly met our criteria was challenging. Therefore, we opted to collect data ourselves, utilizing the website *whoscored.com*. The website provides various information about matches from major football leagues and tournaments worldwide, including detailed descriptions of the events performed by each player on the field. Each match has its own dedicated page, and upon inspecting the HTML source code of these web pages, we confirmed that the information about the events occurring during the match was intricately embedded in that source code in chronological order. Consequently, we created a web crawler to collect the HTML files of each match's webpage and utilized regular expressions to access and extract event information from the source code of these files.

As mentioned, *whoscored.com* offers match information from various leagues and tournaments. However, we specifically focused on the EPL and the English Football League Championship (EFL). These two leagues represent the top two tiers of professional football leagues in England and are among the most prestigious leagues globally, featuring top-level players and managers showcasing matches of the highest caliber. We collected all league matches data from both leagues spanning the 20/21, 21/22, and 22/23 seasons, totaling 2,835 matches. However, during the preprocessing stage, 27 matches were excluded due to data errors. Among the remaining matches, there were 1,200 matches (42.7%) where the home team won, 905 matches (32.2%) where the away team won, and 703 (25.1%) matches that ended in a draw. The total number of successful passes in all matches was 1,728,846, with the home team accounting for 887,025 passes and the away team for 841,821 passes, averaging 615 passes per match.

Data preprocessing

Our preprocessing steps focused on constructing the passing networks as graph structures and extracting non-pass in-game features from the embedded match information data. Compared to traditional approaches that rely on manually extracted passing-related variables, our method directly represents the passing network as a structured graph and processes it using Graph Attention Networks (GATs). This approach ensures that node and edge relationships are dynamically learned rather than predefined through

³ <http://whoscored.com>

handcrafted metrics. This process involved segregating the data into player-centric components and event-specific components. The player-centric component contributed to forming passing graph node features, while the event-specific component facilitated the construction of graph edges and in-game feature vectors. Examples of these datasets are represented in Tables 1 and 2.

The node features, subsequently integrated into graph embeddings, comprised 7 features: a player's position (label-encoded), average X and Y pitch coordinates, height, weight, rating, and passing accuracy. All features except passing accuracy underwent min-max scaling for the entire match, ensuring their values fell within the range of 0 to 1. As shown in Table 3, some of the node features are static, while others are dynamic. The player-centric component contains information on the players' positions, height, and weight, which do not change as the game progresses. Therefore, the values of these features are static. In contrast, ratings, pass accuracy, and average X and Y coordinates are dynamic features, as their values change as the game proceeds. We used the player's

Table 1 An example of player-related data

PlayerId	ShirtNo	Name	Position	Height	Weight	IsFirstEleven	TeamId
331422	99	Mile Svilar	GK	189	77	True	84
136481	19	Zeki Celik	DR	180	78	True	84
244804	23	Gianluca Mancini	DC	190	77	True	84
71345	6	Chris Smalling	DC	194	81	True	84
92329	37	Leonardo Spinazzola	DL	186	75	True	84
93700	16	Leandro Paredes	MC	182	75	True	84
...	...	...	...	...	...	...	...

Table 2 An example of event-related data

PlayerId	EventType	Minute	Second	TeamId	x	y	EndX	EndY	OutcomeType	ReceiverId
425319	Pass	0	23	18	25.9	76.7	6.4	64.8	Successful	384868
384868	Pass	0	27	18	6.9	66.4	26.3	96.6	Successful	331466
331466	Balltouch	0	29	18	29.7	96.9	NaN	NaN	Unsuccessful	255167
255167	Pass	0	44	29	69.9	0.0	72.5	10.4	Unsuccessful	141732
141732	Foul	0	45	18	27.6	90.7	NaN	NaN	Unsuccessful	143595
143595	Foul	0	45	29	72.4	9.4	NaN	NaN	Unsuccessful	331466
331466	Pass	1	0	18	23.2	91.7	19.7	89.7	Successful	251134
...	...	...	...	...	...	...	...	...	...	...

Bold font indicates the highest value of each performance metric

Table 3 An example of graph node features

Player (position)	Position	Height	Weight	Ratings	Pass acc.	X-coord.	Y-coord.
Player 1 (goalkeeper)	12	0.5789	0.5745	0.3476	0.5128	0.0583	0.5491
Player 2 (defender)	3	0.5526	0.1702	0.4028	0.8000	0.3082	0.7106
Player 3 (midfielder)	4	0.7368	0.5745	0.4076	0.8200	0.2727	0.4079
...	...	...	...	...	...	...	...
Player 11 (attacker)	13	0.4474	0.3191	0.4250	0.8400	0.4771	0.5296

rating corresponding to the prediction timeframe since it is updated every minute and recorded in our data. Calculating pass accuracy and X, Y coordinates is straightforward, as shown in Table 2. For pass accuracy, we calculate the proportion of successful passes out of the total passes made up to the prediction timeframe for each player. For the X, Y coordinates, we take the average of the coordinates recorded for each player up to the prediction timeframe. One thing to mention is that we only used the starting eleven players of each team as nodes. Therefore the information for the player nodes who were sent-off or substituted are no longer updated.

For the event-related data, all events that occurred during a match were recorded in chronological order, with each sample representing a single event by a player as shown in Table 2. Each sample contains information such as the type of event, the time it occurred, the player and his team, the starting and ending coordinates of the event, and whether the event was successful or not. The bolder entries are used to illustrate that event data is record chronologically, and that actions from different teams may alternate within the same time frame. Furthermore, both successful and unsuccessful events are equally logged in the dataset. To process this data, we first designated the player of the next sample as a new column called ‘receiver’ for all samples. If the team information of a specific sample and its next sample are different, it means that the next sample corresponds to an event from the opposing team’s player. This automatically signifies that the result of the previous sample was ‘unsuccessful’ and gave the possession of the ball to the opposing team. Therefore, by retaining only the samples of events with successful results, we ensured that the information about the ‘receiver’ was only valid within the same team.

After that, we partitioned the data into pass-event data and non-pass-event data based on the sample’s event type. Pass events were used to construct edges in the passing graph. Given that we could identify both the passer and the receiver of each pass, we generated separate edge index arrays for the passers and receivers of each team. Additionally, we created edge feature arrays containing the count of passes between each pair of elements in the edge index arrays. As previously mentioned, we retained information only for the eleven players who constituted the starting members of the match. Moreover, while creating the node features, we computed their average X and Y coordinates on the pitch in advance. This enabled us to establish edges between nodes based on how frequently each player passed to others from their average coordinates during the match. Even for edges between a pair of nodes, the number of passes might differ depending on whether a player acted as the passer or the receiver. Therefore, our passing graph assumed the form of a directed graph, where the number of passes between players was represented as edge weights. The procedure of creating edges between nodes based on consecutive passes is depicted in Fig. 3.



Fig. 3 An example of a passing network in which nodes and edges are added according to successive passes

On the other hand, non-pass events contributed to the creation of the non-pass in-game features vector $\mathcal{I}_t(H)$ for the home team and $\mathcal{I}_t(A)$ for the away team (t represents the t -th match). Unlike pass events, which provide player-level information, in-game features must represent collective insights for the entire team-level performance. Consequently, we defined 16 indicators for the in-game features vector with team-level information. They include the total counts of ball touches, substitutions, aerial competitions, ball recovery (BR), blocked passes (BP), cards, clearances, corners awarded (CA), disposessions, fouls, interceptions, missed shots (MS), saved shots (SS), tackles, take-ons (TO), and offside provoked (OP). BR and BP refer to the number of times the ball is stolen and the number of times it is stolen from the team, respectively. Moreover, TO refers to the number of attempts to beat the opponent by dribbling, and OP refers to the number of times a player is caught offside, which is an aggressive indicator. To ensure uniformity, both the number of passes and in-game feature values underwent min-max scaling based on the entire match in the training set.

Above features were selected based on their availability in the dataset and their relevance as commonly observed performance indicators during a football match. Several of them, such as missed shots, fouls, and cards, have been used in previous studies focused on match outcome prediction [24, 25]. Importantly, all features are not sourced from past games. This ensures that the model does not rely on outcome-deterministic information such as current score or goals conceded, thereby preserving the predictive objectivity of our real-time framework (Table 4).

Therefore, the obtained passing network graphs and the in-game features vectors for the t -th match can be represented as $\mathcal{G}_t(H) = (\mathcal{V}_t(H), \mathcal{E}_t(H), \mathbf{A}_t(H))$ and $\mathcal{I}_t(H)$ for the home team, and $\mathcal{G}_t(A) = (\mathcal{V}_t(A), \mathcal{E}_t(A), \mathbf{A}_t(A))$ and $\mathcal{I}_t(A)$ for the away team. Here, $\mathcal{V}_t(\cdot)$ is the node set, $\mathcal{E}_t(\cdot)$ is the edge set, and $\mathbf{A}_t(\cdot)$ is the adjacency matrix. Specifically, the node set $\mathcal{V}_t(\cdot)$ consists of $\{v_i(\cdot), \dots, v_N(\cdot)\}_t$ with $N = 11$, and $v_i(\cdot) \in \mathbb{R}^7$ since each team's passing network has 11 players and there are a total of 7 defined node features. As mentioned, we utilized the total number of passes between two players as edge weights. This forms the adjacency matrix $\mathbf{A}_t(\cdot) = a_{ij}(\cdot)_{11 \times 11}$ with its element $a_{ij}(\cdot)$ being either 0 or the total number of passes between players $v_i(\cdot)$ and $v_j(\cdot)$.

Note that we kept using $(\cdot)$ notation to clarify that each graph and its elements (node and edge set, and adjacency matrix), as well as the in-game features vector belong to each team, and therefore one match inevitably yields two passing network graphs and two in-game features vectors.

Model structure

In Fig. 4, we present an outline of the three graph classification models designed for this study. The task we tried to solve in this study is to classify each football match's final outcome into three categories using passing networks and in-game feature vectors for both

Table 4 An example of the employed in-game features

Touch	Sub	Aerial	BR	BP	Card	Clearance	CA	...	Tackle	TO	OP
0.3963	0.4000	0.2879	0.3521	0.3636	0.2857	0.3175	0.0625	...	0.2609	0.1034	0.0000

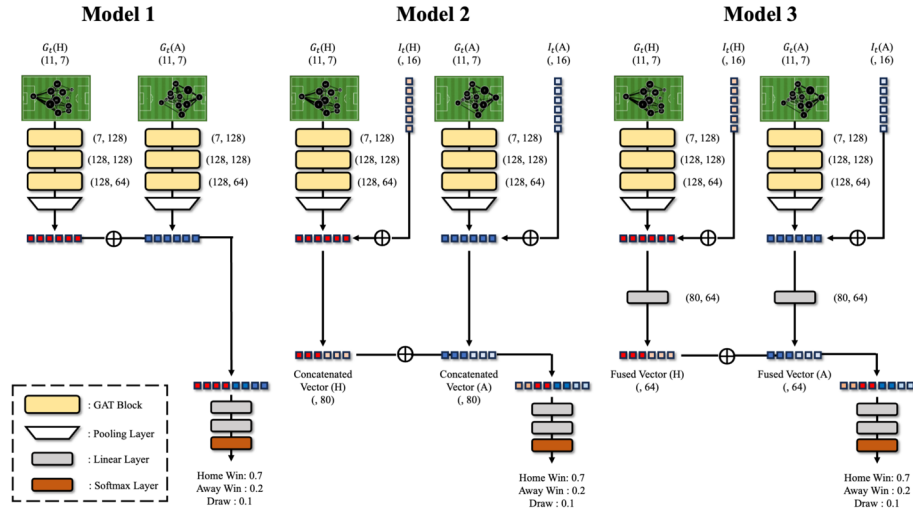


Fig. 4 Graph-based classification model structure for predictive football match outcomes; Model 1 receives only the passing networks as input. Model 2 utilizes in-game features through concatenation with graph embeddings. Model 3 incorporates both graph embeddings and in-game features through vector fusion

the home and the away team participating in that match. Therefore, we can formalize the predicted outcome of the t th match $\hat{Y}_t$ as follows:

$$\hat{Y}_t = \operatorname{argmax}_Y \Pr(Y | \mathcal{G}_t(H), \mathcal{I}_t(H), \mathcal{G}_t(A), \mathcal{I}_t(A)) \quad (1)$$

The three models, Model 1, Model 2, and Model 3, share similarities in terms of their structural aspects, but they exhibit slight differences based on the objectives they aim to address. Let's first examine the commonalities among each of the models.

As can be seen in Fig. 4, both $\mathcal{G}_t(H)$ and $\mathcal{G}_t(A)$ pass through three GAT Blocks consist of the GAT layer, the ELU activation function, and the dropout layer. We can denote the updated node features after first GAT block as

$$\mathbf{X}_t^1(\cdot) = \operatorname{Dropout}(\operatorname{ELU}(\operatorname{GAT}(\mathcal{G}_t(\cdot)))) \quad (2)$$

The GAT layer in equation 2 will first calculate attention coefficients

$$e_{ij}(\cdot) = \operatorname{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}v_i(\cdot) \oplus \mathbf{W}v_j(\cdot)]) \quad (3)$$

for each node v_i and v_j , where $\mathbf{W} \in \mathbb{R}^{F' \times F}$ is the weight matrix for linear transformation with F input dimension ($F = 7$ since this is the first GAT block) and F' output dimension and $\mathbf{a} \in \mathbb{R}^{2F'}$ is the attention mechanism's parameters. Then it will normalize the attention coefficients and yield an attention score using softmax.

$$\alpha_{ij}(\cdot) = \frac{\exp(e_{ij}(\cdot))}{\sum_{k \in \mathcal{N}_i(\cdot)} \exp(e_{ik}(\cdot))} \quad (4)$$

Here, $\mathcal{N}_i(\cdot)$ represents the neighboring nodes set of node $v_i(\cdot)$. The updated node feature $h_i^1(\cdot)$ will be the weighted sum of the neighboring linear transformed node with attention scores as weights. Since we didn't utilize multi-head attention in the experiment, $h_i(\cdot)$ can be represented as

$$h_i^1(\cdot) = \sigma \left(\sum_{j \in \mathcal{N}_i(\cdot)} \alpha_{ij} \mathbf{W} v_j(\cdot) \right) \quad (5)$$

with σ to notate non-linear function. The $h_i^1(\cdot)$ will again go through the second GAT block and be updated again to yield $h_i^2(\cdot)$ and finally, after passing through the third GAT blocks, the three-time updated node features set $\{h_i^3(\cdot), \dots, h_N^3(\cdot)\}$ goes through a graph pooling layer to be transformed into the graph embeddings. We used simple average pooling method to take the average of node embedding.

$$\mathbf{x}_{\mathcal{G}_t(\cdot)} = \frac{1}{|\mathcal{V}_t(\cdot)|} \sum_{i=1}^N h_i^3(\cdot) \quad (6)$$

As mentioned, despite the structural commonalities of the three models, they are different to each other in terms of the way they utilize $\mathcal{I}_t(\cdot)$ as additional inputs. The first model (Model 1) is designed to predict $\hat{Y}_t$ only with $\mathcal{G}_t(\cdot)$. Therefore, the graph embeddings for the home and the away team after the pooling layers are concatenated together to make a combined vector for the match $\mathbf{x}_{\mathcal{G}_t} = \mathbf{x}_{\mathcal{G}_t(H)} \oplus \mathbf{x}_{\mathcal{G}_t(A)}$ and this goes towards the final softmax layer to output three predicted probabilities for each class. On the other hand, the second model (Model 2) and the third model (Model 3) aim to improve the predictive performance from Model 1 by incorporating $\mathcal{I}_t(\cdot)$, providing richer information about the ongoing match. Therefore, they receive total four inputs; $\mathcal{G}_t(H)$, $(I)_t(H)$, $\mathcal{G}_t(A)$, and $\mathcal{I}_t(A)$. What's different is the way they combine the information from each input. Model 2 combines information from $\mathcal{G}_t(\cdot)$ and $\mathcal{I}_t(\cdot)$ through simple concatenation. Therefore, the combined vector for the match is represented as

$$\mathbf{x}_{\mathcal{G}_t} = \{\mathbf{x}_{\mathcal{G}_t(H)} \oplus \mathcal{I}_t(H)\} \oplus \{\mathbf{x}_{\mathcal{G}_t(A)} \oplus \mathcal{I}_t(A)\} \quad (7)$$

However, the third model (Model 3) incorporated a fusion technique building on Gallo et al's work [42] to fuse the graph embeddings and the in-game features vector. They are first concatenated to each other as in Model 2, but the concatenated vectors of both teams are not just concatenated again but each of them goes through an additional linear layer, merging the insights derived from the graph and the non-pass events information. This makes the combined vector for the match of Model 3, which is represented as:

$$\mathbf{x}_{\mathcal{G}_t} = \text{Linear}(\{\mathbf{x}_{\mathcal{G}_t(H)} \oplus \mathcal{I}_t(H)\}) \oplus \text{Linear}(\{\mathbf{x}_{\mathcal{G}_t(A)} \oplus \mathcal{I}_t(A)\}) \quad (8)$$

This combined vector progresses through two additional linear layers and finally goes through a softmax layer, which yields the probability of categorization into one of the three classes.

We chose to use GAT layer over other graph neural network architectures due to the unique structural properties of passing interactions. Unlike social or citation networks with thousands of nodes, football passing networks are relatively small-scale, typically consisting of only 11 nodes per team, yet they exhibit highly dynamic and structured interactions. Traditional GNNs such as Graph Convolutional Networks (GCNs) aggregate information equally from neighboring nodes, which may not be optimal when certain passes hold significantly more importance than others in defining team strategies.

Additionally, compared to Graph Isomorphism Networks (GINs) or message-passing-based architectures, GATs enhance interpretability by making explicit which passes contribute most to network centrality and overall team cohesion. Given that passing networks are inherently sparse, yet highly structured in terms of connectivity and role-specific importance, the ability of GATs to selectively prioritize influential nodes and edges makes them the optimal choice for modeling team coordination and match prediction in football analytics.

Empirical analysis

This study aims to predict the final outcome of ongoing football matches based on the continuously updated passing graphs during the match. Therefore, we conducted experiments to compare the predictive performance at the 45 th, 60 th, 75 th, and 90 th minute of the match, using the passing networks and non-pass events information available at that exact time point. By comparing the results of Models 1, 2, and 3 simultaneously, we analyzed the contributions of the passing graphs, in-game features, and the fusion technique to the model's predictions. Furthermore, by testing the model to predict the outcomes of the top five European leagues except the EPL, we showed that our proposed graph classification model operates effectively regardless of the targeted leagues.

For the experiments, we partitioned the entire dataset into training (1684 samples), validation, and test sets (562 samples each). Given the smaller sample size for the Draw class in comparison to other classes, we set class weights accordingly. The model's hyperparameters were configured as follows: a batch size of 64, a dropout rate of 0.5, 1000 epochs with the patience of 50 for early stopping, and a learning rate of 0.001 reducing by 1% per epoch using a scheduler [43–46]. A batch size of 16 was used for the passing graphs of the 45 th minute.

We compared the performance of the proposed models by inputting the passing networks and in-game features vector generated at each time point during the match (45 th, 60 th, 70 th, and 90 th minutes). Here, by simultaneously comparing the three models, we examined how the differences in each model affect its predictive performance. Additionally, we conducted experiments to evaluate how well our model performs in predicting match outcomes in the top 5 European leagues to demonstrate that the graph-based method proposed in this study is not only applicable to the English league but also effective for other major European football leagues. For these experiments, we applied the same procedures for the 60 th, 70 th, and 90 th minutes. However, a batch size of 16 was used for the passing graphs of the 45 th minute since it demonstrated better performance in a grid search. During the training process, we set the loss function as cross-entropy and utilized the Adam optimizer to update the parameters of the neural network. Plus, we configured early stopping to prevent overfitting.

Evaluation metrics

To evaluate the model performance, we report four standard classification metrics: *Precision*, *Recall*, *F1-score*, and *Accuracy*. These metrics are widely used in machine

Table 5 The performance of the models at the 45-minute marks

Model	Category	Precision	Recall	F1-score	Accuracy	Baseline	F1-score
Model 1	Home Win	85.53%	54.17%	66.33%	55.66%	RF	59.59%
	Away Win	58.48%	60.24%	59.35%			37.50%
	Draw	0.00%	0.00%	0.00%			18.60%
Model 2	Home Win	85.09%	56.23%	67.71%	56.60%	XGBoost	60.13%
	Away Win	61.40%	61.40%	61.40%			43.45%
	Draw	0.76%	7.14%	1.38%			30.57%
Model 3	Home Win	85.96%	55.21%	67.24%	57.36%	SVM	62.03%
	Away Win	59.65%	65.38%	62.39%			40.51%
	Draw	4.58%	31.58%	8.00%			12.99%

Bold font indicates the highest value of each performance metric

Table 6 The performance of the models at the 60-minute marks

Model	Category	Precision	Recall	F1-score	Accuracy	Baseline	F1-score
Model 1	Home Win	86.61%	61.98%	72.25%	61.58%	RF	60.13%
	Away Win	71.67%	63.55%	67.36%			43.45%
	Draw	5.07%	35.00%	8.86%			10.23%
Model 2	Home Win	88.70%	62.72%	73.48%	62.66%	XGBoost	58.25%
	Away Win	72.22%	65.00%	68.42%			45.81%
	Draw	5.07%	36.84%	8.92%			26.73%
Model 3	Home Win	87.45%	65.51%	74.91%	63.56%	SVM	58.47%
	Away Win	76.67%	62.73%	69.00%			46.77%
	Draw	5.07%	38.88%	8.97%			6.58%

Bold font indicates the highest value of each performance metric

learning and correspond to commonly known statistical terms such as Positive Predictive Value, Sensitivity, and Balanced Accuracy.

- *Precision* (also known as Positive Predictive Value) measures the proportion of correctly predicted positive instances among all predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- *Recall* (also referred to as Sensitivity or True Positive Rate) measures the proportion of correctly predicted positive instances among all actual positives:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- *F1-score* is the harmonic mean of Precision and Recall, providing a balanced measure especially when classes are imbalanced:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Table 7 The performance of the models at the 75-minute marks

Model	Category	Precision	Recall	F1-score	Accuracy	Baseline	F1-score
Model 1	Home Win	85.83%	74.64%	79.84%	70.41%	RF	65.85%
	Away Win	76.80%	72.02%	74.33%			56.02%
	Draw	35.71%	54.35%	43.10%			28.87%
Model 2	Home Win	86.25%	74.19%	79.77%	70.59%	XGBoost	63.20%
	Away Win	75.69%	73.66%	74.66%			54.02%
	Draw	37.14%	54.17%	44.07%			36.86%
Model 3	Home Win	86.25%	79.92%	82.97%	75.22%	SVM	65.01%
	Away Win	79.56%	78.69%	79.12%			55.11%
	Draw	50.71%	59.66%	54.83%			25.12%

Bold font indicates the highest value of each performance metric

Table 8 The performance of the models at the 90-minute marks

Model	Category	Precision	Recall	F1-score	Accuracy	Baseline	F1-score
Model 1	Home Win	85.83%	84.43%	85.12%	77.58%	RF	74.11%
	Away Win	82.32%	84.18%	83.23%			68.46%
	Draw	57.45%	57.45%	57.45%			40.74%
Model 2	Home Win	86.25%	85.54%	85.89%	79.00%	XGBoost	69.82%
	Away Win	84.53%	84.53%	84.53%			64.21%
	Draw	59.57%	60.43%	60.00%			39.67%
Model 3	Home Win	88.95%	86.78%	87.14%	83.09%	SVM	74.62%
	Away Win	88.95%	89.94%	89.45%			66.49%
	Draw	68.09%	68.09%	68.09%			44.25%

Bold font indicates the highest value of each performance metric

- *Accuracy* measures the proportion of all correct predictions (both positive and negative) over the total number of instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Here, *TP*, *TN*, *FP*, and *FN* denote true positives, true negatives, false positives, and false negatives, respectively. While we focus on these four metrics, we acknowledge that other measures such as specificity and the area under the ROC curve (AUC-ROC) are also widely used in statistical evaluation and may be explored in future work.

Performance comparison at different time points

Tables 5, 6, 7, and 8 present the classification performance of the three models for the Home Win, Away Win, and Draw classes at each time point. Additionally, the performance of three baseline models-Random Forest, XGBoost, and SVM Classifier-is presented on the right side of the corresponding tables. These baseline models used the in-game features of both teams, which are also used as inputs for Model 2 and Model 3. In addition to this, passing information was aggregated into total pass count and average pass accuracy features to be used as additional features for the baselines. Our proposed graph-based models that directly utilize the passing network outperformed the baseline

models across all time points and classes, except for the draw class at the 45-minute and 60-minute marks. Although the baseline models demonstrated better classification ability for the draw class at these specific time points, the superior performance of our models in other classes underscores the advantage of using the passing network. This confirms that, as explained in the introduction, employing the passing network, which includes information about the locations where passes occur and the players who make them, rather than relying solely on aggregated pass information, enables the model to better understand the dynamics of the match and extract the necessary information to predict the final outcome more effectively.

As expected, the overall performance of all three models improved as the match progressed towards its conclusion since both passing networks and in-game features contained richer information about the match as it unfolded. However, one thing to highlight is that our models demonstrated meaningful performance even when a significant portion of the match remained. In Table 5, Although there were no accurately predicted samples for the Draw class, Model 1 exhibited F1-Scores of approximately 66% and 59% for the Home Win and Away Win classes, respectively. The performance of Model 1 improved significantly as the match progressed, demonstrating remarkable performance with F1-Scores of 85% for the Home Win, 83% for the Away Win, and 57% for the Draw class when utilizing passing networks containing information up to the 90th minute of the match as in Table 8. Given that no information was provided regarding non-pass events, the result suggests that the passing event itself contains more decisive information than initially expected to predict the final outcome of the match.

The performance of Model 2 and Model 3, which utilized information from the non-pass features alongside the passing networks, was also a subject of interest. Comparing the performance of Model 1, Model 2, and Model 3 across Tables 5 to 8, we observed that the provision of non-pass events information can achieve improvement in model performance, but how this information is intermingled with graph embeddings played a more significant role. Upon comparing the results of Model 1 and Model 2, the performance of Model 2 was higher compared to that of Model 1. However, examining the experimental results of Model 3, which used the idea of vector fusion to mix the two vectors with a fully connected layer, we observed much higher classification performance across all classes and time points, especially in the Draw class. In our designed model, graph embeddings were first player-level node information and later transformed into team-level information after passing through three GAT blocks and a pooling layer. On the other hand, the in-game features vector was already a team-level vector, aggregated by summing up non-pass events occurrences. Therefore, rather than simply concatenating these two vectors, using even a single linear layer to facilitate the fusion of the two vectors is more effective in connecting player-level information with team-level information and ultimately achieving superior model performance.

Additionally, to compare the performance of our proposed methodology for predicting the outcome of an ongoing match with the methodologies from past studies that predict the outcome of a match that has not yet started, we compared the results of our models with those reported in one specific previous study. It might seem intuitive that using information generated during a match to predict the outcome of an ongoing match would naturally perform better than using past match data to predict the

outcome before the match begins. However, these past match data contains crucial and decisive information about the performance of the two teams, such as their head-to-head records, goals scored and conceded in previous matches, current earned points, and even recent win rates. Previous studies have actively utilized these metrics for their input variables. In contrast, our models rely solely on information gained during the current match and, in the case of Model 1, only use the passing network. Therefore, we conducted this comparison to demonstrate that it is possible to achieve sufficiently good performance in predicting ongoing match outcomes with perfect avoidance of any information from past matches. Table 9 presents the baseline experimental results from the study by Rodriguez and Pinto [24]. We chose their approach as our only baseline model because, like ours, they trained and tested their model on data from the EPL matches and provided evaluation metrics for each class that aligned with our research. Furthermore, their study employed a by-example approach of utilizing pre-match information and machine learning models to predict the upcoming match's outcomes. This facilitated a direct comparison of the two methodologies: using pre-match data to predict the outcomes before the match versus using ongoing match data to predict the outcomes during the match. Our experiment, utilizing only passing networks up to the 60 th minute, yielded F1 scores for the Home Win and Away Win classes similar to the baseline. However, when incorporating information up to the 75 th minute, our model exhibited superior performance across all classes in every evaluation metric.

This experiment reveals another significant finding: while the in-game features vector does contain information about various non-pass events, how it is combined with information from passing networks significantly impacts the model's predictability. In the experiment using data up to the 60 th minute, Model 2 exhibited lower predictive performance compared to Model 1, which does not utilize the in-game features vector at all. In the experiment using data after the 75 th minute, although the performance of Model 2 was superior to that of Model 1, the increased gap was smaller than expected. Conversely, Model 3, which integrates the in-game features vector with the graph embeddings through fusion techniques, demonstrated the highest F1-Score and accuracy for all classes except for the 60 th minute. The graph embeddings can be considered as team-level information integrated from individual players (nodes) and their passing (edges) information after being processed through the GAT blocks and a pooling layer. However, since the information from the in-game features vector is already at the team level, simply concatenating these two vectors and feeding them to the model may not be as effective as expected. Even a single hidden layer, such as a linear layer, that combines the two pieces of information can be more effective in improving predictive performance. Furthermore, features such as the number of shots, offside occurrences, and substitutions in

Table 9 The baseline research's predictive performance for the same three categories of football match outcome

Researcher	Category	Precision	Recall	F1-score	Accuracy
Rodriguez and Pinto [24]	Home Win	68.47%	81.29%	74.51%	65.26%
	Away Win	65.74%	65.14%	65.46%	
	Draw	50.00%	29.76%	37.30%	

Table 10 The performance of Model 3 on other leagues at the 45-minute marks

League	Category	Precision	Recall	F1-score	Accuracy
EPL and EFL championship	Home win	85.96%	55.21%	67.24%	57.36%
	Away win	59.65%	65.38%	62.39%	
	Draw	4.58%	31.58%	8.00%	
Serie A	Home win	88.82%	58.19%	67.83%	59.72%
	Away win	67.83%	67.83%	67.83%	
	Draw	2.15%	15.38%	3.77%	
La Liga	Home win	85.47%	61.76%	71.71%	61.19%
	Away win	59.72%	58.33%	59.72%	
	Draw	7.69%	85.71%	14.12%	
Bundesliga	Home win	83.97%	62.38%	71.58%	58.46%
	Away win	55.91%	51.49%	53.61%	
	Draw	9.21%	50.00%	15.56%	
Ligue 1	Home win	80.26%	61.62%	69.71%	58.99%
	Away win	68.97%	55.56%	61.54%	
	Draw	9.09%	57.14%	15.69%	

Bold font indicates the highest value of each performance metric

Table 11 The performance of Model 3 on other leagues at the 90-minute marks

League	Category	Precision	Recall	F1-score	Accuracy
EPL and EFL championship	Home win	88.95%	86.78%	87.14%	83.09%
	Away Win	88.95%	89.94%	89.45%	
	Draw	68.09%	68.09%	68.09%	
Serie A	Home win	92.02%	91.46%	91.74%	84.38%
	Away Win	84.55%	85.95%	85.25%	
	Draw	71.43%	70.71%	71.07%	
La Liga	Home win	87.78%	89.77%	88.76%	81.72%
	Away Win	82.24%	88.89%	85.44%	
	Draw	68.24%	59.79%	63.74%	
Bundesliga	Home win	91.36%	90.80%	91.08%	84.32%
	Away win	91.58%	82.08%	86.57%	
	Draw	61.73	72.46%	66.67%	
Ligue 1	Home win	90.06%	94.16%	92.06%	84.64%
	Away win	83.33%	87.72%	85.47%	
	Draw	76.67%	66.99%	71.50%	

Bold font indicates the highest value of each performance metric

the in-game features vector have a relatively greater impact on draw results than on win or lose outcomes. This allowed the F1-Score for the Draw class to increase more steeply than for other classes when the in-game features vector was utilized with fusion.

Cross-league prediction comparison

The second experiment aimed to assess how well our model predicts the outcomes of the match from the top 5 European leagues outside of the EPL. This is to ensure that the model has not overfitted to the matches from the English league during training, thereby compromising its robustness on outside-of-the-English-league matches. We collected additional match data from Serie A (Italy), La Liga (Spain), Bundesliga (Germany),

and Ligue 1 (France) for the 22/23 season. Then, utilizing passing networks and in-game features vectors at the 45-minute and 90-minute marks, we employed Model 3 to predict the outcomes of each match and compared the results with those of the same model at the same time points.

Tables 10 and 11 compare the performance of Model 3 for each league at the 45 th minute and 90 th minute marks, respectively. The results for the EPL and the EFL Championship at the top of both tables are consistent with the Model 3 results in Tables 5 and 8. The numbers highlighted in bold in each table indicate cases where the performance exceeded that for the EPL and EFL Championship matches. For example, in Table 10 the precision score for the Away Win class of Serie A matches is written in bold because it outperformed that of EPL and EFL Championship matches by 5.44%p.

From Tables 10 and 11, we can see that, despite our model being trained exclusively on English league data, it maintained consistent classification performance and even recorded better metrics when predicting outcomes in new leagues it had not encountered during training. Particularly in Table 10, the draw class showed significant improvements in La Liga, the Bundesliga, and Ligue 1. In Table 11, which presents predictions at the 90-minute mark, there were no leagues where the F1 score for the home team win class was lower than that of the test set. The overall accuracy was also comparable to or higher than that of the test set when predictions were made at both the 45-minute and 90-minute marks.

The graph embeddings

As validated by the two preceding experiments, the prediction of the final outcome of ongoing football matches utilizing the passing graph has exhibited promising predictive efficacy across all outcome classes. Therefore, further ablation study was taken to analyze how our model effectively generates graph embeddings from the passing networks to capture the dynamics of each participating team in the match.

Upset matches

To evaluate how effectively the graph embeddings capture the distinctions between teams, we gathered “upset matches” from the ongoing 23/24 EPL season, where the outcomes deviated from general expectations. These expectations were gathered from pre-match 3-way betting odds provided by *whoscore.com*. In 3-way betting, bettors can select one of three possible outcomes in a football match: Home Win, Draw, and Away Win. We defined “upset matches” as games where the team predicted to win based on pre-match betting odds ultimately lost instead (excluding matches where the actual outcome was a draw). Filtering out 40 matches from the 23/24 EPL season that met this criterion, we inputted the model with the 45-minute passing network graphs to compare the model’s predictions with the actual outcomes of these upset matches. Remarkably, using only the passing network graphs from the first half, the model accurately predicted 75% (30 out of 40) of the upset matches’ outcomes.

To investigate the factors contributing to the model’s high predictive accuracy even in cases where match outcomes deviate from common expectations, we extracted the graph embeddings of the upset matches with correct predictions from the model. The

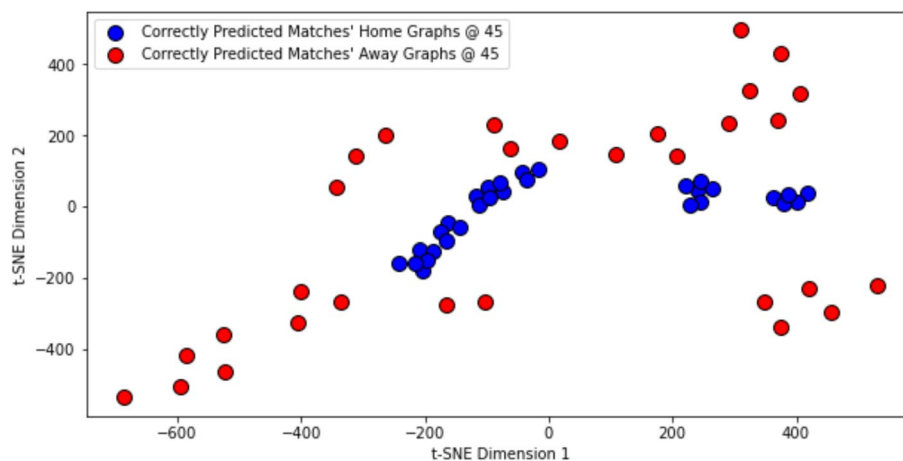


Fig. 5 t-SNE visualization of correctly predicted matches' home and away graph embeddings at 45 th minute

extraction was made when both the home and away team's graphs passed through the model's pooling layer, resulting in 64-dimensional team-level vectors. Subsequently, we visualized the embeddings of the home and away teams' graphs using t-SNE dimension reduction [47]. As depicted in Fig. 5, the graph embeddings of the home team and the away team form distinct clusters, effectively distinguishing each other. Consequently, the model successfully differentiated between the two graphs and predicted the match outcome accordingly.

This analysis reveals that the passing network in football matches contains rich information, even within the first 45 min alone. Moreover, our proposed approach in this study, which involves substituting the passing network with graphs and converting them into graph embeddings, effectively captures the characteristics of these graphs and appropriately positions them within the vector space.

Then, we computed the average cosine similarity between the graph vectors of the home/away teams in matches where the model accurately predicted the outcome and, conversely, between the graph vectors of the home/away teams in matches where the prediction failed. For the 45-minute graph, the average cosine similarity for the former was -0.039 , while for the latter, it was 0.342 . As for the 90-minute graph embeddings, the former yielded -0.023 , whereas the latter resulted in 0.9 . This indicates that the upset matches with accurately predicted outcomes have very low similarity between the passing graphs of the home and away teams. For a more intuitive understanding,

To showcase our model's proficiency in properly mapping graph embeddings from the passing graphs to the vector space, we conducted an intriguing experiment: comparing the passing graphs of the top 2 teams and the bottom 2 teams of the ongoing 23/24 EPL season. Before analysis, we posited that the passing graphs would encapsulate features reflecting each team's performance throughout the season and that distinguishable characteristics would emerge between the top-ranked and bottom-ranked teams. As of the last week of March 24 in the EPL, Arsenal ranked first, followed by Liverpool in second place, while Burnley and Sheffield United occupied the 19 th and 20 th positions,

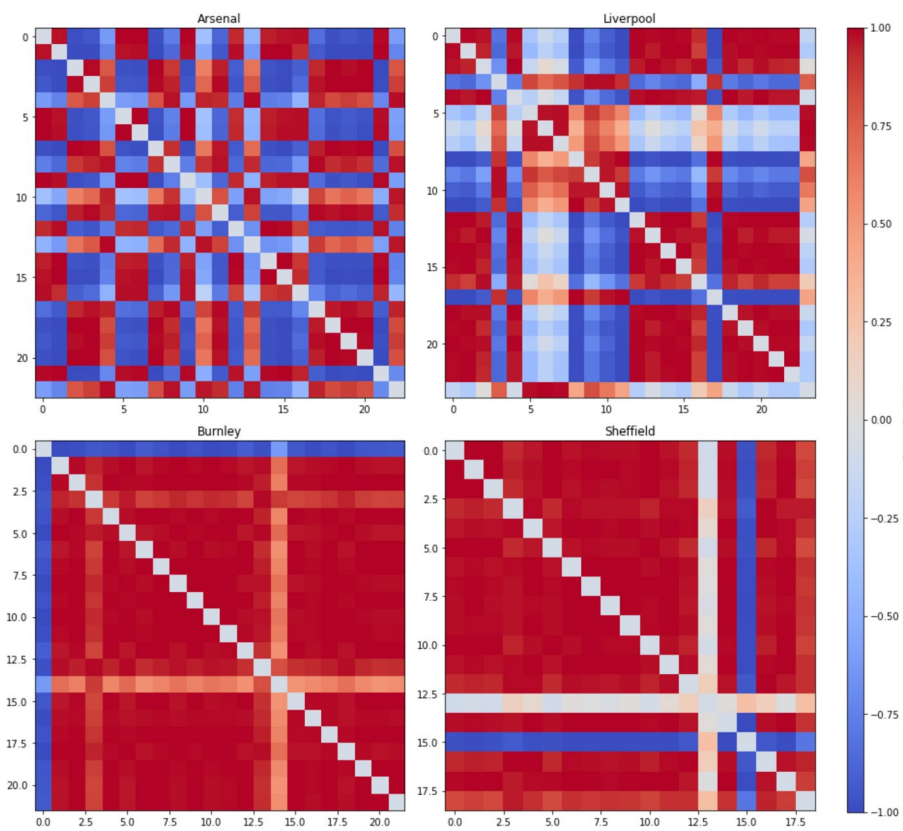


Fig. 6 Cosine similarities heatmap of the top 2 and the bottom 2 teams from the 23/24 EPL (orthogonal elements are erased)

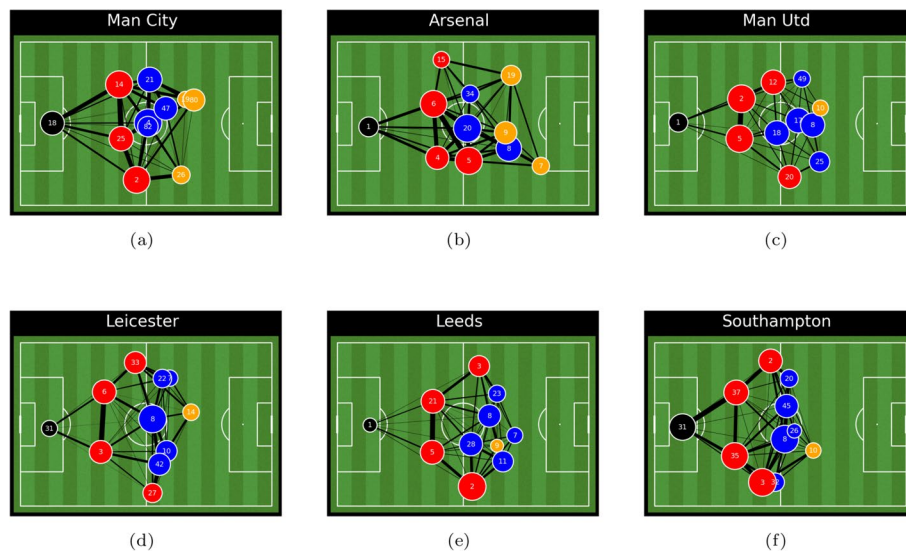
respectively. We collected data from matches involving these four teams and retained only the graphs of matches where our model accurately predicted the outcomes.

Figure 6 illustrates a heatmap visualizing the cosine similarities between the graph embeddings of each team's matches. For Arsenal and Liverpool, the cosine similarity oscillates between values close to 1 or close to -1 . In contrast, Burnley and Sheffield United exhibit cosine similarities close to 1 for almost all matches. This suggests that the top two teams have distinct passing networks drawn for each match, resulting in embedding vectors with opposite or orthogonal directions in the vector space. Conversely, the bottom two teams have passing networks of similar shapes, leading to embedding vectors pointing in similar directions for each match.

The fact that the tactical diversity between top-tier and lower-tier teams can be clearly distinguished through graph embeddings indicates that our model effectively learns to position the dynamic characteristics of each team within the vector space by representing their passing networks. Moreover, the ability to leverage these differences represented in the graph embeddings enables our model to accurately predict the outcome of a match. Plus, the distinctiveness of passing graphs between top and bottom teams, as well as the discernibility of these differences in the graph embeddings, suggest the potential for further research utilizing passing graphs and graph modeling to capture various dynamics of a football match. This finding aligns with

Table 12 The Top three and the bottom three teams with their last home win match for the EPL 22/23 season

Season	Teams	Rank	Match
22/23	Manchester City	1	May 21, 2023 (vs. Chelsea)
	Arsenal	2	May 28, 2021 (vs. Wolves)
	Manchester United	3	May 28, 2023 (vs. Fulham)
	Leicester City	18	May 28, 2023 (vs. Westham)
	Leeds United	19	April 4, 2023 (vs. Nottingham Forest)
	Southampton	20	March 4, 2023 (vs. Leicester City)

**Fig. 7** Passing network of the top three/bottom three teams' last home win match during the EPL 22/23 season: (a) Manchester City, (b) Arsenal, (c) Manchester United, (d) Leicester City, (e) Leeds United, (f) Southampton

previous studies that have used network-based methods to characterize playing styles in football. For example, Diquigiovanni and Scarpa [48] used clustering methods on passing networks to identify team styles, while Palazzo et al. [49] analyzed triadic patterns to distinguish tactical preferences across teams. Our study builds upon these approaches by demonstrating that graph embeddings can encapsulate the dynamic characteristics of team tactics.

The experiment affirms that our proposed graph classification model can predict the overall game outcome with a relatively high F1 score using the passing network and other indicators generated within the first half of the game. This underscores the significance of passing, the fundamental movement of football, which contains more game-related information than initially presumed.

The top three teams and the bottom three teams

A thorough analysis was conducted to compare the passing networks of the top three and bottom three teams in the EPL during the 22/23 season. The objective was to discern the extent to which the passing graph reflects tactical disparities between these

teams and how these variations correlate with their final rankings. To achieve this, the passing networks of the top three and bottom three teams were meticulously visualized, with a specific focus on the final home victory game of the season. The top three and bottom three teams along with each team's final home victory of the season, are meticulously outlined in Table 12. Furthermore, Fig. 7 provides a visual representation of each team's passing network during their last home win game. Notably, players' positions are distinguished by color, with defenders represented in red, midfielders in blue, and attackers in yellow.

The significant differences observed in the passing networks of the top three teams and the bottom three teams, as depicted in Fig. 5 are outlined as follows:

- The positioning of fullbacks
- The presence of a player distributing the ball in the midfield

The integration of fullbacks into the midfield zone signifies a tactical approach that can profoundly impact a team's strengths and weaknesses. As discussed in the article by [50], the EPL 22/23 season emphasized the strategic use of fullbacks in the midfield to exert dominance over control and facilitate a playmaker's freedom to create chances. This strategy is prominently evident in the passing networks of the top three teams, presenting a distinct contrast to the playing styles of the bottom three teams. The passing networks of the top three teams showcase a dynamic utilization of fullbacks, highlighting their involvement in the midfield to create numerical advantages and contribute to ball circulation. Conversely, in the passing networks of the bottom three teams, a noticeable pattern emerges wherein the fullbacks predominantly occupy both flanks of the pitch. This pattern suggests a reduced level of involvement in the midfield compared to the fullbacks of the top three teams, indicating a challenge in effectively adopting the prevailing trend of fullbacks penetrating the midfield—a tactical nuance that defined the dynamics of the 22/23 season.

Furthermore, the presence of players in the midfield who act as connectors between the defensive and offensive lines is a critical aspect of a team's passing strategy. It is intriguing to observe how the passing networks of the top three teams exhibit a more cohesive and interconnected structure in the midfield, facilitating smoother ball circulation. The crucial consideration lies in whether the positions of the players tasked with ball distribution effectively maintain equilibrium between the defensive and offensive lines. In the case of the top three teams, these players strategically position themselves at the precise center of the two lines, significantly contributing to the build-up phase and mitigating the risk of a pronounced gap emerging between the defensive and offensive segments.

In contrast, a distinctive pattern emerges for the bottom three teams. While the midfielders naturally occupy the central area of the field, their positioning tends to be relatively closer to the attacking line. Consequently, they face challenges in seamlessly participating in the build-up process from the defensive line to the offensive line. This strategic disparity underscores a noteworthy distinction in the orchestration of play between the top and bottom-performing teams, particularly in their ability to synchronize defensive and offensive transitions through the central midfield.

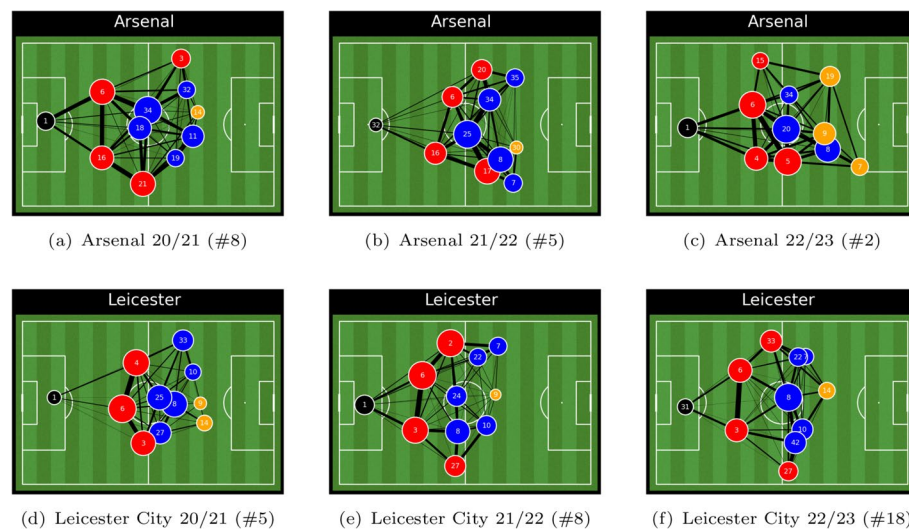


Fig. 8 Passing network of each team's last home win match per season

Table 13 Teams that have ascended/declined in rankings for 3 consecutive seasons and each team's last home win match per season

Teams	Season	Rank	Match
Arsenal	20/21	8	May 23, 2021 (vs. Brighton)
	21/22	5	May 22, 2022 (vs. Everton)
	22/23	2	May 28, 2023 (vs. Wolves)
Leicester City	20/21	5	April 26, 2021 (vs. Crystal Palace)
	21/22	8	May 22, 2022 (vs. Southampton)
	22/23	18	May 28, 2023 (vs. Westham)

Teams with rising/falling in rankings for consecutive seasons

The information regarding teams that experienced consistent season-to-season ranking ascension or decline and each team's last home win game per each season, determined as subjects for analysis in the final section, is summarized in Table 13. Furthermore, the passing networks of each game can be examined in Fig. 8.

The observed changes within each team, unfolding throughout the seasons, manifest in the form of the passing network. For Arsenal, the passing network is transitioned from a horizontal structure to a vertical one. In contrast, for Leicester City, the passing network, which was initially vertical, transforms into a horizontal configuration. In other words, from the players' perspective, Arsenal has evolved from a style characterized by horizontally wide but shallow passes to one where vertical, deep passes are exchanged. Conversely, Leicester City has transitioned from delivering vertical, deep passes to a style featuring horizontally wide but shallow passes. A vertical pass ultimately represents an attempt to move the ball towards the opponent's territory. Regardless of its diagonal or vertical nature, such forward passes are far more effective in leading to specific sequences, culminating in actions like shots, compared to lateral passes or backward passes [51]. Furthermore, as the season progressed, Arsenal witnessed a tightening of spaces between players, leading to a blur in the distinction between positions, thereby

becoming less constrained to specific locations. In contrast, Leicester City's passing network exhibited a clearer distinction in positions, and the spacing between lines expanded over time.

Through detailed analysis of the passing networks for each team in these two cases, we were able to discern differences in passing networks between teams that achieved successful season outcomes and those with less favorable performances. Additionally, we observed changes in the passing networks of teams that consistently improved or worsened their season performances. However, since our analysis in the appendix focused solely on the last home victory game of each team, expanding the scope to include all games throughout the entire season for further research may reveal a more systematic and explicit relationship between season performance and the patterns in passing networks. We anticipate that such an expanded analysis could provide clearer insights into the dynamics between overall season outcomes and the characteristics of passing networks.

Conclusion

In this study, we proposed a novel framework for real-time football match outcome prediction by modeling passing networks as dynamic graphs and processing them with graph-based classification models. Our approach departs from traditional feature engineering by capturing the structural dynamics of team interactions in real time. We introduced three model variations, each incorporating increasing levels of in-game context.

Our results demonstrate that prediction accuracy consistently improves as matches progress, with the best-performing model achieving high accuracy across all outcome classes near full time. Notably, the model maintains strong performance across multiple European leagues, suggesting its generalizability beyond specific competitions. We also showed that the graph embeddings learned by our models reflect team-specific tactical patterns, offering interpretability and insight into match dynamics.

However, challenges remain-particularly in early-match prediction and for the Draw class, which exhibits greater outcome uncertainty. Improving early-stage predictions may require more expressive representations or additional contextual features.

A potential direction for future research lies in integrating historical performance data with real-time passing network analysis. While historical context can enrich predictions, it also risks reducing model adaptability. Hybrid models that dynamically balance short-term and long-term information-such as temporal graph networks or attention-based fusion mechanisms-may offer a path forward. We hope this work provides a foundation for further exploration of structure-aware, interpretable, and real-time predictive models in football analytics.⁴

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Author contributions

JL and EP designed this research. JL collected the datasets. JL and EP analyzed and interpreted the collected datasets. JL and EP wrote the manuscript. EP and APdP supervised the research and revised the manuscript. All authors read and approved the final manuscript.

⁴ <https://anonymous.4open.science/r/We-Know-Who-Wins-F3C7/>

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Data availability

All codes are available at <https://github.com/dxlabssku/We-Know-Who-Wins>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Tokuyama S, Greenwell TC. Examining similarities and differences in consumer motivation for playing and watching soccer. *Sport Mark Q*. 2011;20(3):148–56.
2. Martins AM, Cró S. The demand for football in Portugal: new insights on outcome uncertainty. *J Sports Econ*. 2018;19(4):473–97.
3. Joseph A, Fenton NE, Neil M. Predicting football results using Bayesian nets and other machine learning techniques. *Knowl Based Syst*. 2006;19(7):544–53.
4. Hucaljuk J, Rakipović A. Predicting football scores using machine learning techniques. In: 2011 Proceedings of the 34th International Convention MIPRO, IEEE; 2011. p. 1623–1627.
5. Baboota R, Kaur H. Predictive analysis and modelling football results using machine learning approach for English premier league. *Int J Forecast*. 2019;35(2):741–55.
6. Cintia P, Rinivillo S, Pappalardo L. A network-based approach to evaluate the performance of football teams. In: *Machine Learning and Data Mining for Sports Analytics Workshop*, Porto, Portugal. 2015.
7. López Peña J, Touchette H. A network theory analysis of football strategies. *arXiv arXiv:1206.6904*, preprint. 2012.
8. Paixão P, Sampaio J, Almeida CH, Duarte R. How does match status affects the passing sequences of top-level European soccer teams? *Int J Perform Anal Sport*. 2015;15(1):229–40.
9. Rein R, Raabe D, Memmert D. 'Which pass is better?' novel approaches to assess passing effectiveness in elite soccer. *Hum Mov Sci*. 2017;55:172–81.
10. Buldú JM, Busquets J, Martínez JH, Herrera-Diestra JL, Echegoyen I, Galeano J, Luque J. Using network science to analyse football passing networks: dynamics, space, time, and the multilayer nature of the game. *Front Psychol*. 2018;9:1900.
11. Kawasaki T, Sakaue K, Matsubara R, Ishizaki S. Football pass network based on the measurement of player position by using network theory and clustering. *Int J Perform Anal Sport*. 2019;19(3):381–92.
12. Pappalardo L, Cintia P. Quantifying the relation between performance and success in soccer. *Adv Complex Syst*. 2018;21(03n04): 1750014.
13. Bilek G, Ulas E. Predicting match outcome according to the quality of opponent in the English premier league using situational variables and team performance indicators. *Int J Perform Anal Sport*. 2019;19(6):930–41.
14. AlMulla J, Islam MT, Al-Absi HR, Alam T. Soccernet: a gated recurrent unit-based model to predict soccer match winners. *PLoS ONE*. 2023;18(8):0288933.
15. Veličković P, Cucurull G, Casanova A, Romero A, Liò P, Bengio Y. Graph attention networks. *Proc ICLR'18*. 2018;2018:1–12.
16. Beheshtian-Ardakani A, Salehi M, Sharma R. CMPN: modeling and analysis of soccer teams using complex multiplex passing network. *Chaos Solitons Fractals*. 2023;174: 113778.
17. Rahimian P, Kim H, Schmid M, Toka L. Pass receiver and outcome prediction in soccer using temporal graph networks. In: *International Workshop on Machine Learning and Data Mining for Sports Analytics*. Cham: Springer;2023. p. 52–63.
18. Dixon MJ, Coles SG. Modelling association football scores and inefficiencies in the football betting market. *J Roy Stat Soc C Appl Stat*. 1997;46(2):265–80.
19. Rue H, Salvesen Ø. Focus on sport: prediction and retrospective analysis of soccer matches in a league. *J Royal Statist Soc Ser D Statist*. 2000;49(3):399–418.
20. Goddard J. Regression models for forecasting goals and match results in association football. *Int J Forecast*. 2005;21(2):331–40.
21. Hvattum LM, Arntzen H. Using ELO ratings for match result prediction in association football. *Int J Forecast*. 2010;26(3):460–70.

22. Carpita M, Sandri M, Simonetto A, Zuccolotto P. Discovering the drivers of football match outcomes with data mining. *Quality Technol Quantitat Manag*. 2015;12(4):561–77.
23. Wheatcroft E. Forecasting football matches by predicting match statistics. *J Sports Analytics*. 2021;7(2):77–97.
24. Rodrigues F, Pinto A. Prediction of football match results with machine learning. *Proc Comput Sci*. 2022;204:463–70.
25. Stübinger J, Mangold B, Knoll J. Machine learning in football betting: prediction of match results based on player characteristics. *Appl Sci*. 2019;10(1):46.
26. Rahman MA. A deep learning framework for football match prediction. *SN Appl Sci*. 2020;2(2):165.
27. Merhej C, Beal R.J, Matthews T, Ramchurn S. What happened next? using deep learning to value defensive actions in football event-data. In: *Proceedings of KDD '21, Association for Computing Machinery, New York, NY*; 2021. p. 3394–3403.
28. Fenil E, Manogaran G, Vivekananda G, Thanjaivadivel T, Jeeva S, Ahilan A, et al. Real time violence detection framework for football stadium comprising of big data analysis and deep learning through bidirectional lstm. *Comput Netw*. 2019;151:191–200.
29. Cuperman R, Jansen KM, Ciszewski MG. An end-to-end deep learning pipeline for football activity recognition based on wearable acceleration sensors. *Sensors*. 2022;22(4):1347.
30. Carpita M, Ciavolino E, Pasca P. Players' role-based performance composite indicators of soccer teams: a statistical perspective. *Soc Indic Res*. 2021;156:815–30.
31. Cefis M, Carpita M. Accuracy and explainability of statistical and machine learning xG models in football. *Statistics*. 2024;59(1):1–20.
32. Grund TU. Network structure and team performance: the case of English premier league soccer teams. *Soc Netw*. 2012;34(4):682–90.
33. Yamamoto Y, Yokoyama K. Common and unique network dynamics in football games. *PLoS ONE*. 2011;6(12):29638.
34. Buldu J, Busquets J, Echegoyen I, Seirul. Io F. Defining a historic football team: using network science to analyze Guardiola's FC Barcelona. *Sci Rep*. 2019;9(1): 13602.
35. Mclean S, Salmon PM, Gorman AD, Stevens NJ, Solomon C. A social network analysis of the goal scoring passing networks of the 2016 European football championships. *Hum Mov Sci*. 2018;57:400–8.
36. Levoli R, Palazzo L, Ragozini G. On the use of passing network indicators to predict football outcomes. *Knowl-Based Syst*. 2021;222: 106997.
37. Gama J, Passos P, Davids K, Relvas H, Ribeiro J, Vaz V, Dias G. Network analysis and intra-team activity in attacking phases of professional football. *Int J Perform Anal Sport*. 2014;14(3):692–708.
38. Gonçalves B, Coutinho D, Santos S, Lago-Penas C, Jiménez S, Sampaio J. Exploring team passing networks and player movement dynamics in youth association football. *PLoS ONE*. 2017;12(1):0171156.
39. Cartas A, Ballester C, Haro G. A graph-based method for soccer action spotting using unsupervised player classification. In: *Proceedings of ACM MMSports '22*, 2022:93–102.
40. Zhou W, Yu G, You S, Wang Z. An improved passing network for evaluating football team performance. *Appl Sci*. 2023;13(2):845.
41. Cao S. Passing path predicts shooting outcome in football. *Sci Rep*. 2024;14(1):9572.
42. Gallo I, Ria G, Landro N, La Grassa R. Image and text fusion for UPMC food-101 using bert and cnns. In: *2020 35th International Conference on Image and Vision Computing New Zealand (IVCNZ)*, IEEE; 2020. p. 1–6.
43. Kim J, Ahn H, Park E. Multi-pop: enhancing user engagement with content-based multimodal popularity prediction in social media. *Expert Syst*. 2024;41(12):13707.
44. Lee M, Ji H, Park E. Deepaup: a deep neural network framework for abnormal underground heat transport pipelines. *IEEE Trans Autom Sci Eng*. 2023;21(2):2017–26.
45. Kim S, Park E. STAD-GCN: spatial-temporal attention-based dynamic graph convolutional network for retail market price prediction. *Expert Syst Appl*. 2024;255: 124553.
46. Park E. CRNet: a multimodal deep convolutional neural network for customer revisit prediction. *J Big Data*. 2023;10(1):1.
47. Maaten L, Hinton G. Visualizing data using t-SNE. *J Mach Learn Res*. 2008;9(11):2579–605.
48. Diquigiovanni J, Scarpa B. Analysis of association football playing styles: an innovative method to cluster networks. *Stat Model*. 2019;19(1):28–54.
49. Palazzo L, Levoli R, Ragozini G. Testing styles of play using triad census distribution: an application to men's football. *J Quant Anal Sports*. 2023;19(2):125–51.
50. Tharm L. Premier league 2022-23 tactical trends: Defenders in midfield, Back Fours and through balls 2023. <https://theathletic.com/4559625/2023/05/31/premier-league-tactical-trends/>.
51. Cordon-Carmona A, Garcia-Aliaga A, Marquina M, Calvo JL, Mon-Lopez D, Refoyo Roman I. What is the relevance in the passing action between the passer and the receiver in soccer? study of elite soccer in la liga. *Int J Environ Res Public Health*. 2020;17(24):9396.

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